KAYA VISION

Vision Point to Vision Point II

Migration Guide

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20283 State Road 7, Suite 350
Boca Raton, FL 33498
USA
+1 (561) 689-2899

info@kaya.vision
support@kaya.vision

www.kaya.vision

+1 (561) 689-2899
info@kaya.vision
www.kaya.vision

1 Revision History

Version	Date	Notes
2025.1.0	12.2024	Initial release
2025.1.1-1.2	01-03.2025	The document remains unchanged
2025.1.3	04.2025	A set of functions have been implemented in the Adapter library (see Table 3 – Implemented functions)
2025.2.0	06.2025	KYFG_StreamCreateAndAlloc, KYFG_StreamLinkFramesContinuously have been implemented in the Adapter library
2025.2.1-2026.1.0	12.2025-01.2026	The document remains unchanged
2026.1.1	04.2026	KYDeviceEventCallBackRegister, KYDeviceEventCallBackUnregister have been implemented in the Adapter library
2026.1.2	05.2026	The document remains unchanged
2026.1.3	10.2026	KYFG_StreamGetAux, KYFG_BufferAnnounceChunks have been implemented in the Adapter library

Table 1 – Revision History

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4 Introduction

4.1 Safety precautions

Please take the time to read through the precautions listed below to prevent preventable and unnecessary injuries and damage to you, other personnel, or property. Read these safety instructions carefully before your first use of the product, as these precautions contain safety instructions that must be observed. Be sure to follow this manual to prevent misuse of the product.

 Caution! Read Carefully and do not disregard these instructions.
In the event of a failure, disconnect the power supply. Disconnect the power supply immediately and contact our sales personnel for repair. Continuing to use the product in this state may result in a fire or electric shock.
If an unpleasant smell or smoking occurs, disconnect the power supply. Disconnect the power supply immediately! Continuing to use the product in this state may result in a fire or electric shock. After verifying that no smoking is observed, contact our sales personnel for repair.
Do not disassemble, repair or modify the product. This may result in a fire or electric shock due to a circuit shortage or heat generation. Contact our sales personnel before inspection, modification or repair.
Do not place the product on unstable surfaces. Otherwise, it may drop or fall, resulting in injury to persons or the camera.
Do not use the product if dropped or damaged. Otherwise, a fire or electric shock may occur.
Do not touch the product with metallic objects. Otherwise, a fire or electric shock may occur.
Do not place the product in dusty or humid environments, nor where water may splash. Otherwise, a fire or electric shock may occur.
Do not wet the product or touch it with wet hands. Otherwise, the product may fail or it may cause a fire, smoking or electric shock.
Do not touch the gold-plated sections of the connectors on the product. Otherwise, the surface of the connector may be contaminated by sweat or skin-oil, resulting in contact failure of a connector, malfunction, fire or electric shock due to static electricity discharge.
Do not use or place the product in the following locations. ▪ Unventilated areas such as closets or bookshelves. ▪ Near oils, smoke or steam. ▪ Next to heat sources. ▪ A closed (and not running) car where the temperature becomes high. ▪ Static electricity replete locations ▪ Near water or chemicals. Otherwise, a fire, electric shock, accident or deformation may occur due to a short circuit or heat generation.
Do not place heavy objects on the product. Otherwise, the product may be damaged.
Be sure to discharge static electricity from the body before touching any sensitive electronic components. The electronic circuits in the computer and the circuits on the board are sensitive to static electricity and surges. Improper handling may seriously damage the circuits. In addition, do not let your clothing come in contact with the circuit boards or components. Otherwise, the product may be damaged.

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4.2 Disclaimer

KAYA Vision will assume no responsibility for any damage that may ensue by the use of this product for any purpose other than intended, as previously stated. Without detracting from what was previously written, please be advised that the company will take no responsibility for any damages caused by:

- Earthquake, thunderstrike, natural disasters, fire caused by use beyond our control, wilful and/or accidental misuse and/or use under other abnormal and/or unreasonable conditions.
- Secondary damages caused by the use of this product or its unusable state (business interruption or others).
- Use of this product in any manner that contradicts this manual or malfunctions that may occur due to connection to other devices. Damage to this product that is out of our control or failure due to modification
- Accidents and/or third parties that may be involved.

Additionally, KAYA Vision assumes no responsibility or liability for:

- Erasure or corruption of data caused by the use of this product.
- Any consequences or other abnormalities following the use of this product

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5 Overview

The purpose of this document is to list and demonstrate the ways to migrate to the new Vision Point II API.

This API is to be used with KAYA Frame Grabbers hardware provided by KAYA Vision. This is a high-level API for connecting, configuring and capturing data streaming over 1, 2, 4 or 8 channels. KAYA Frame Grabbers are capable of connecting to various cameras at various speeds and topologies.

5.1 Document Structure

This migration guide is divided into few major topics each related to different functionalities:

- Legacy Vision Point API Adapter.
- Transition to the new API.

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6 Legacy Vision Point API Adapter

Use the new possibilities with Vision Point II, knowing that existing code is supported. To ensure a smooth transition, a legacy Vision Point API Adapter was developed. The Adapter allows to use the Vision Point II libraries with existing code developed using the legacy Vision Point API.

6.1 Old code usage

To enable the customer to use the Vision Point API Adapter, it is necessary to link to the Adapter library.

1. Install Vision Point II from the official source.
2. Open an existing project in Visual Studio (or another preferred application).
3. Add `KAYA_VISION_POINT_2_LIB_PATH` to the /LIBPATH.
4. Link to the special Adapter library.

Vision Point API library	Vision Point II API Adapter library
KYFGLib.lib	KYFGLibA_vc141.lib

Table 2 – Adapter library

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6.2 Implemented functions

The most of Vision Point function were implemented to the Adapter library. The full list is below:

Function name	implemented
Basic flow functions	
KY_GetSoftwareVersion	Since 2024.1.0
KYFGLib_Initialize	Since 2024.1.0
KY_DeviceScan	Since 2024.1.0
KY_DeviceInfo	Since 2024.1.0
KYFG_OpenEx	Since 2024.1.0
KYFG_Open	Since 2024.1.0
KYFG_UpdateCameraList	Since 2024.1.0
KYFG_CameraScanEx	Since 2024.1.0
KYFG_CameraInfo2	Since 2024.1.0
KYFG_CameraOpen2	Since 2024.1.0
KYFG_CameraStart	Since 2024.1.0
KYFG_CameraStop	Since 2024.1.0
KYFG_CameraClose	Since 2024.1.0
KYFG_Close	Since 2024.1.0
Stream Interface	
KYFG_StreamGetInfo	Since 2024.1.0
KYFG_StreamGetSize	Since 2024.1.0
KYFG_StreamGetFrameIndex	Since 2024.1.0
KYFG_StreamGetPtr	Since 2024.1.0
KYFG_StreamCreate	Since 2024.1.0
KYFG_StreamCreateAndAlloc	Since 2025.2.0
KYFG_StreamBufferCallbackRegister	Since 2024.1.0
KYFG_StreamBufferCallbackUnregister	Since 2024.1.0
KYFG_StreamDelete	Since 2024.1.0
Buffer Interface	
KYFG_BufferGetInfo	Since 2024.1.0
KYFG_BufferRevoke	Since 2024.1.0
KYFG_BufferAllocAndAnnounce	Since 2024.1.0
KYFG_BufferAnnounce	Since 2024.1.0
KYFG_BufferToQueue	Since 2024.1.0
KYFG_BufferQueueAll	Since 2024.1.0
Grabber Parameters	
KYFG_SetGrabberValue	Since 2024.1.0
KYFG_SetGrabberValueInt	Since 2024.1.0
KYFG_SetGrabberValueFloat	Since 2024.1.0
KYFG_SetGrabberValueBool	Since 2024.1.0
KYFG_SetGrabberValueEnum	Since 2024.1.0
KYFG_SetGrabberValueString	Since 2024.1.0
KYFG_GrabberExecuteCommand	Since 2024.1.0
KYFG_SetGrabberValueEnum_ByValueName	Since 2024.1.0
KYFG_GrabberWriteReg	Since 2024.1.0
KYFG_DeviceDirectHardwareWrite	Since 2024.1.0
KY_RegisterGrabberConfigurationParameterCallback	Since 2024.1.0
KY_UnregisterGrabberConfigurationParameterCallback	Since 2024.1.0

KYFG_GetGrabberValue	Since 2024.1.0
KYFG_GetGrabberValueInt	Since 2024.1.0
KYFG_GetGrabberValueEnum	Since 2024.1.0
KYFG_GetGrabberValueFloat	Since 2024.1.0
KYFG_GetGrabberValueBool	Since 2024.1.0
KYFG_GetGrabberValueStringCopy	Since 2024.1.0
KYFG_GetGrabberValueIntMaxMin	Since 2024.1.0
KYFG_GetGrabberValueFloatMaxMin	Since 2024.1.0
KYFG_GetGrabberValueType	Since 2024.1.0
KYFG_IsGrabberValueImplemented	Since 2024.1.0
KYFG_GetGrabberValueRegister	Since 2024.1.0
KYFG_GrabberReadReg	Since 2024.1.0
KYFG_DeviceDirectHardwareRead	Since 2024.1.0
KY_GetGrabberConfigurationParameterDefinitions	Since 2024.1.0
KY_GetGrabberPropertyParameterValue	Since 2024.1.2
KYFG_ReadPortReg	Since 2025.1.2
KYFG_ReadPortBlock	Since 2025.1.2
KYFG_WritePortReg	Since 2025.1.2
KYFG_WritePortBlock	Since 2025.1.2
KYFG_GetPortStatus	Since 2025.1.2
KYFG_StreamLinkFramesContinuously	Since 2025.2.0
KYFG_AuxDataCallbackRegister	Since 2025.1.3
KYFG_AuxDataCallbackUnregister	Since 2025.1.3
KYFG_StreamGetAux	Since 2026.1.3
KYFG_BufferAnnounceChunks	Since 2026.1.3
KYFG_BufferSubmit	No
KYFG_LoadPatternData	No
KYFG_LoadFileData	No
KYFG_CameraSendEventMessage	No
KYFG_DevicePortSendEventMessage	No
Camera Parameters	
KYFG_SetCameraValueInt	Since 2024.1.0
KYFG_SetCameraValueFloat	Since 2024.1.0
KYFG_SetCameraValueBool	Since 2024.1.0
KYFG_SetCameraValueEnum	Since 2024.1.0
KYFG_SetCameraValueString	Since 2024.1.0
KYFG_CameraExecuteCommand	Since 2024.1.0
KYFG_SetCameraValue	Since 2024.1.0
KYFG_SetCameraValueEnum_ByValueName	Since 2024.1.0
KYFG_CameraWriteReg	Since 2024.1.0
KY_RegisterCameraConfigurationParameterCallback	Since 2024.1.0
KY_UnregisterCameraConfigurationParameterCallback	Since 2024.1.0
KYFG_CameraGetXML	Since 2024.1.0
KYFG_GetCameraValueInt	Since 2024.1.0
KYFG_GetCameraValueEnum	Since 2024.1.0
KYFG_GetCameraValueFloat	Since 2024.1.0
KYFG_GetCameraValueBool	Since 2024.1.0
KYFG_GetCameraValue	Since 2024.1.0
KYFG_IsCameraValueImplemented	Since 2024.1.0
KYFG_GetCameraValueType	Since 2024.1.0

KYFG_GetCameraValueStringCopy	Since 2024.1.0
KYFG_GetCameraValueIntMaxMin	Since 2024.1.0
KYFG_GetCameraValueFloatMaxMin	Since 2024.1.0
KYFG_CameraReadReg	Since 2024.1.0
KY_GetCameraConfigurationParameterDefinitions	Since 2024.1.0
KYFG_GetCameraValueRegister	Since 2024.1.0
KY_GetCameraPropertyParameterValue	Since 2024.1.2
KYFG_CameraCallbackRegister (please use KYFG_StreamBufferCallbackRegister)	No
KYFG_CameraCallbackUnregister (please use KYFG_StreamBufferCallbackUnregister)	No
KYDeviceEventCallBackRegister	Since 2026.1.1
KYDeviceEventCallBackUnregister	Since 2026.1.1
Camera Simulator Parameters	
KYCS_InjectImageHeader	No
KYCS_InjectVideoCRCErrors	No
KYCS_InjectControlCRCErrors	No
KYCS_ReadBootstrapRegs	No
KYCS_WriteBootstrapRegs	No
KYCS_GetImageHeader	No
KYCS_GenerateCxpEvent	No
Firmware update	
KYFG_LoadFirmware	Since 2024.1.2
KYFG_GetFirmareUpdateFileInfo	Since 2024.1.2

Table 3 – Implemented functions

In case of using not implemented function the user will receive a callback message “Function is not implemented”. More functions might be added to the Adapter in next Vision Point II releases.

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6.3 Transition to the new API

When working with the Adapter as described above, it translates legacy API calls to the corresponding new API. It does so by creating new handles internally. To allow a gradual transition from legacy code to the new API, the Adapter implements additional public functions, listed below and declarated in `KYFGLibA.h`.

Important:
After including KYFGLibA.h, make sure to add the include directory `KAYA_VISION_POINT_INCLUDE_PATH` to your project's include paths.

These functions allow the code to retrieve the new API handles used by the Adapter to implement legacy functionality. For instance, if you have a FGHANDLE returned by the legacy API, you can retrieve the new `KYVP_PCI_INTERFACE_HANDLE` associated with it.

- `KYFG_Grabber_Get_KYVP_PCI_INTERFACE_HANDLE()`
  Returns underlying `KYVP_PCI_INTERFACE_HANDLE` for an open FGHANDLE;
  Return value:
  FGSTATUS - Status and error report.

- `KYFG_Grabber_Get_KYVP_COLLECTION_HANDLE()`
  Returns underlying `KYVP_COLLECTION_HANDLE` for an open FGHANDLE
  Return value:
  FGSTATUS - Status and error report.

- `KYFG_Camera_Get_KYVP_DEVICE_HANDLE()`
  Returns underlying `KYVP_DEVICE_HANDLE` for an open FGHANDLE
  Return value:
  FGSTATUS - Status and error report.

REFERENCES

Supported vision standards:

CoaXPress GEN<i>CAM

Vision Point documentation:

Vision Point

TECHNICAL SUPPORT AND PROFESSIONAL SERVICE

If you searched the documents and could not find the answers you need, contact KAYA Vision support service:

- Create a support request on the web: support.kaya.vision
- Our knowledge base is available on: kb.kaya.vision

Visit us at www.kaya.vision for comprehensive information.

SUBMITTING A SUPPORT REQUEST

When opening a support request, please provide the following information when applicable:

For Frame Grabbers:	For Camera:	For Range Extender:
• Vision Point Diagnostic Info* • Serial number of Frame Grabber • Camera model • SFP+ module model • CoaXPress/Fiber cable model and length • External power or PoCXP • PC motherboard model *In the Vision Point app, use menu option Help > Collect diagnostic info.	• Vision Point Diagnostic Info (or frame grabber being utilized) • Serial Number of Camera • XML dump and/or description of how the camera is being utilized • Description of issue • SFP+ module model • CoaXPress/Fiber cable model and length • External power or PoCXP	• Range Extender Model • Serial Number of Range Extender • SFP+ module model • CoaXPress/Fiber Cable model and length • PC configuration • Operating System • Software Revision • Camera and Frame Grabber Manufacturer and Model

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KAYA Vision, Inc.
20283 State Road 7
Suite 350
Boca Raton, FL 33498
USA
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